INTERNATIONAL SEARCH REPORT

International application No.
PCT/US2018/041152

A. CLASSIFICATION OF SUBJECT MATTER

IPC(8) - A61F 2/06; A61F 2/07; A61F 2/82; A61F 2/86; A61F 2/88; A61F 2/89; A61F 2/90 (2018.01)

CPC - A61F 2002/077; A61F 2002/065; A61F 2002/067; A61F 2/82; A61F 2/86; A61F 2/88; A61F 2/89; A61F 2/91 (2018.08)

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

See Search History document

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

USPC - 623/1.1; 623/1.11; 623/1.12; 623/1.13; 623/1.17; 623/1.21; 623/1.22; 623/1.23; 623/1.27; 623/1.35 (keyword delimited)

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

See Search History document

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X --- Y	US 2013/0103135 A1 (TRIVASCULAR, INC.) 25 April 2013 (25.04.2013) entire document	1, 2, 4-6, 9, 10, 15-17, 47 --- 3, 7, 8, 11, 13, 14, 18-32, 36, 38, 39
X --- Y	US 2005/0010277 A1 (CHUTER) 13 January 2005 (13.01.2005) entire document	33-35, 40 --- 36-39
Y	US 2001/0047199 A1 (WIJAY) 29 November 2001 (29.11.2001) entire document	3
Y	US 2016/0000589 A1 (MEDTRONIC VASCULAR, INC.) 07 January 2016 (07.01.2016) entire document	7, 8
Y	US 2010/0161025 A1 (KUPPURATHANAM et al) 24 June 2010 (24.06.2010) entire document	11
Y	US 2013/0296998 A1 (LEOTTA et al) 07 November 2013 (07.11.2013) entire document	13, 14
Y	US 2017/0007263 A1 (ENDOLOGIX, INC.) 12 January 2017 (12.01.2017) entire document	18-32, 37
A	US 2014/0296964 A1 (COOK MEDICAL TECHNOLOGIES LLC) 02 October 2014 (02.10.2014) entire document	1-47
A	US 2015/0157448 A1 (SANFORD HEALTH) 11 June 2015 (11.06.2015) entire document	1-47

☒ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

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"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
19 August 2018	11 SEP 2018
Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US, Commissioner for Patents P.O. Box 1450, Alexandria, VA 22313-1450 Facsimile No. 571-273-8300	Authorized officer Blaine R. Copenheaver PCT Helpdesk: 571-272-4300 PCT OSP: 571-272-7774

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C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 2012/0041544 A1 (WOLF) 16 February 2012 (16.02.2012) entire document	1-47